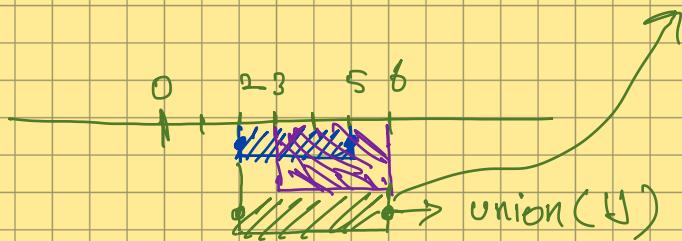
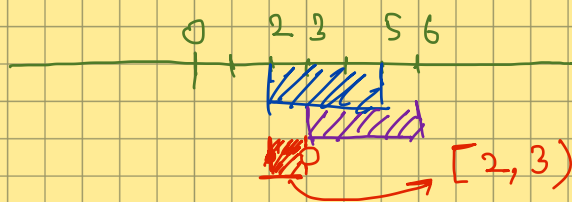


recall!  $[2, 5] = \{x \in \mathbb{R} : 2 \leq x \leq 5\}$

ex:-  $[2, 5] \cup [3, 6] = ? \Rightarrow [2, 6]$



ex:-  $[2, 5] - [3, 6] = ? [2, 3)$



ex:- let  $A = \{(x, y^2) : x \in \mathbb{R}\}$

$B = \{(x, x+2) : x \in \mathbb{R}\}$

Note

$y = x^2$

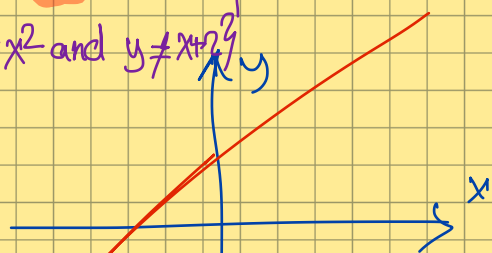
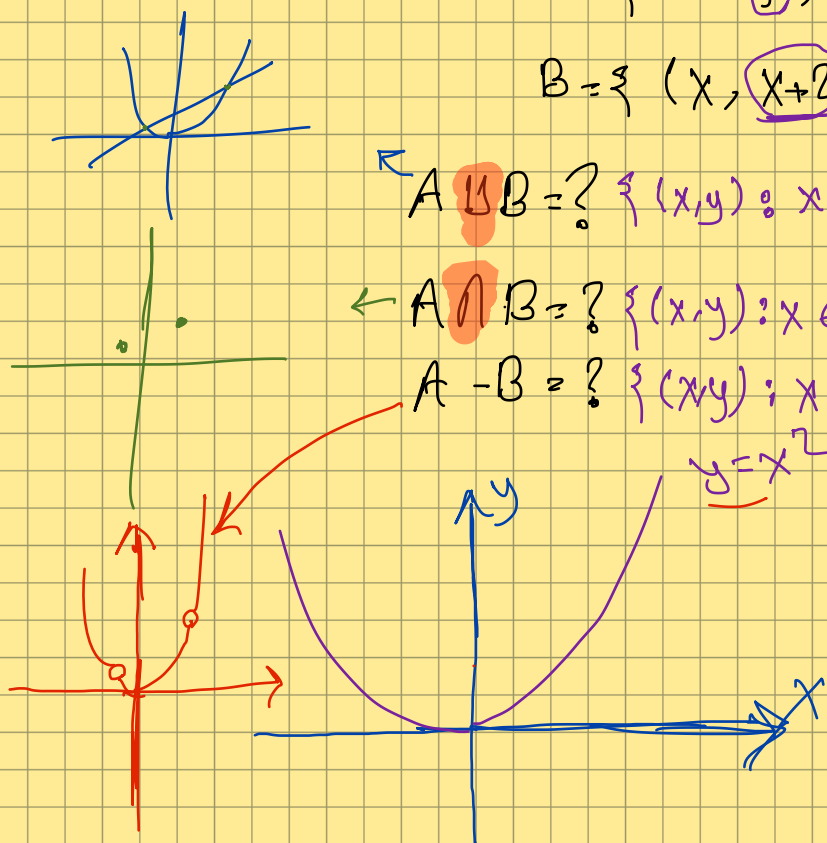
$y = x+2$

$A \cup B = ? \{(x, y) : x \in \mathbb{R}, y = x^2 \text{ or } y = x+2\}$

$A \cap B = ? \{(x, y) : x \in \mathbb{R}, y = x^2 \text{ and } y = x+2\}$

$A - B = ? \{(x, y) : x \in \mathbb{R}, y = x^2 \text{ and } y \neq x+2\}$

$y = x+2$



$\{(x, y) : x \in \mathbb{R}, y = x+2 \text{ and } y \neq x^2\}$

$B - A = y = x+2 \text{ and } y \neq x^2$

